Online controller selection for the angular control of gravitational-wave detectors under non-stationary noise: a multi-armed bandit approach

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Abstract

The angular control loops of gravitational-wave (GW) detectors must reject radiation-pressure (Sidles–Sigg) torques whose strength, together with the seismic and sensing-noise environment, varies over time. In current operation the choice of control filter is changed by hand by operators, for instance switching from low- to high-gain alignment control when an earthquake early-warning system forecasts elevated ground motion. We formulate this choice as an *online controller-selection* problem and study whether a multi-armed bandit policy can perform the selection autonomously so as to minimise a long-horizon cost. Using a time-domain simulator of the pitch degree of freedom of a coupled-cavity detector, validated against a reference implementation, we define three operating regimes and a family of fixed linear controllers with a designed one-to-one regime–controller correspondence, and a scalar reward built from the hard-eigenmode error and control effort. We make three contributions ahead of the bandit study itself. First, we characterise the reward landscape and show that the designed one-to-one regime–controller mapping holds in all three regimes, robustly in the nominal and intermediate regimes and marginally in the extreme regime, with a non-saturating reward. Second, we quantify the cost of switching controllers and find a pronounced asymmetry: gain-decreasing switches are far costlier than gain-increasing ones. Third, and most importantly, we show that the apparent switching penalty is dominated by the *controller-handover implementation*: a naive state reset or raw state carry-over produces large transients, whereas a bumpless, parallel-bank handover renders the switching cost negligible for this gain-scaled controller family. **[TODO: Add one or two sentences summarising the bandit-comparison results (AdSwitch vs. alternatives) once available.]**

1 Introduction

Second-generation gravitational-wave detectors such as Advanced LIGO [1] and Advanced Virgo [2] are kilometre-scale Fabry–Perot–Michelson interferometers whose test masses are suspended as multi-stage pendula. Beyond the longitudinal degrees of freedom used to read out the GW signal, the *angular* degrees of freedom of the mirrors must be actively controlled: residual angular motion couples to the longitudinal signal and, more fundamentally, the circulating optical power exerts radiation-pressure torques that soften one angular eigenmode and stiffen the other—the Sidles–Sigg effect [3]. The alignment sensing and control (ASC) loops that stabilise these modes operate close to a power-dependent instability and are a recognised limit to detector duty cycle and low-frequency sensitivity [4].

The disturbance environment of these loops is strongly non-stationary. Ground motion exhibits a diurnal cycle, a microseismic band that is modulated by weather and oceanic activity on timescales of hours to days, and rare, intense transients associated with earthquakes and storms. A single fixed controller cannot be simultaneously optimal across this range of conditions: a high-gain loop that rejects large low-frequency disturbances injects excess control noise when the environment is quiet, while a low-gain loop that is quiet in nominal conditions fails to maintain alignment during elevated ground motion. In current practice this trade-off is managed manually: operators (or simple rule-based logic) switch between predefined control filters, for example moving to a high-gain alignment configuration when an earthquake early-warning system predicts high incoming ground motion.

This manual, forecast-driven switching is a natural candidate for automation. We pose it as an *online controller-selection* problem: given a fixed library of controllers and a scalar performance signal computed online from the loop's own error and control signals, an autonomous policy must repeatedly choose which controller to apply so as to minimise the time-integrated cost, without prior knowledge of when the environment will change. This is precisely the setting of the *non-stationary multi-armed bandit* [5, 6, 7]: each controller is an arm, the negative cost is a stochastic reward whose distribution shifts when the environment changes, and the policy must balance exploration (testing controllers to detect such shifts) against exploitation (using the controller currently believed best).

In this work we build the simulation and problem infrastructure required to study this question rigorously, and we report a set of results that are prerequisites to—and partly independent of—the choice of bandit algorithm. Our contributions are:

1. A time-domain detector model with an explicit, validated plant, in which three operating regimes and a family of fixed linear controllers are defined so as to admit a designed one-to-one regime–controller mapping (Sections 3–4).

2. A characterisation of the resulting reward landscape (Section 5), establishing that the one-to-one mapping holds in all three regimes (robustly in the nominal and intermediate regimes, marginally in the extreme regime) and that the reward is non-saturating, so a learnable signal is present in every regime.

3. A quantitative study of the *dynamics of switching* (Section 6): the transient cost of changing controllers, its asymmetry in the direction of the gain change, and—central to the feasibility of online switching—the dependence of this cost on the controller-handover implementation, with the conclusion that a realistic bumpless handover makes switching nearly free.

The online-selection experiments that build on this infrastructure are described in Section 7; **[TODO: state whether they are included in this version or deferred]**.

2 Background

2.1 Angular control and the Sidles–Sigg effect

Consider the two test masses forming a single arm cavity of length L with radii of curvature R_1 (input, ITM) and R_2 (end, ETM). In the pitch (vertical-plane tilt) degree of freedom, the cavity geometry is described by the g-factors $g_i = 1 - L/R_i$. The radiation pressure of the circulating power couples the two mirror tilts; diagonalising the coupling yields two angular eigenmodes, conventionally termed *soft* and *hard*, with optical-spring-modified torsional stiffnesses. The transformation from the local (ITM, ETM) tilt basis to the eigenbasis is

$$r = \tfrac{1}{2}\left(g_1 - g_2 + \sqrt{(g_1 - g_2)^2 + 4}\right), \qquad T_{\text{loc}\to\text{eig}} = \frac{1}{1 + r^2}\begin{pmatrix} 1 & r \\ -r & 1 \end{pmatrix}, \tag{1}$$

and the beam-spot-to-angle response factors of the two modes are

$$\left.\frac{\partial y}{\partial \theta}\right|_{\text{soft/hard}} = \frac{L}{2}\,\frac{(g_1+g_2) \pm \sqrt{(g_2-g_1)^2+4}}{g_1 g_2 - 1}. \tag{2}$$

For the parameters of interest the hard mode is the limiting loop: it carries the dominant radiation-pressure torque and is the relevant control-noise injection path, and we therefore use it as the sole performance channel throughout (Section 4). The circulating power that sets the strength of this coupling follows the Fabry–Perot resonance condition

$$P_{\text{cav}} = \frac{P_{\text{in}}\,T_1}{\left|\,1 - \sqrt{R_1 R_2}\,\exp\!\left(4\pi i\,\delta L_{\text{hp}}/\lambda\right)\right|^2}, \tag{3}$$

with T_1 the ITM power transmissivity, λ the laser wavelength, and δL_{hp} the high-passed cavity-length change induced by the mirror tilts.

2.2 The non-stationary disturbance environment

The seismic input to the suspensions varies on multiple timescales: a diurnal component, a microseismic band modulated by weather over hours to days, and sporadic high-amplitude transients. Operationally, this motivates a small set of qualitatively distinct *regimes* and a corresponding set of controllers tuned for each, between which the configuration is switched. A key methodological point, which we adopt throughout, is that physically relevant non-stationarity is *adiabatic*: the environment evolves continuously rather than by abrupt steps, so that the interesting question is whether control remains stable, and a selection policy beneficial, *through* the slow transitions [10]. **[TODO: cite the LIGO/MIT operational practice of earthquake-triggered low/high-gain ASC switching with a proper reference.]**

2.3 Bandits for non-stationary rewards

In the stochastic multi-armed bandit, a policy repeatedly selects one of K arms and observes a reward drawn from that arm's (unknown) distribution, with the goal of minimising regret relative to the best arm [5]. When the reward distributions change over time the relevant notion is *non-stationary* regret against the best arm *at each time*. Two broad families address this: passively forgetting methods, such as discounted or sliding-window UCB [6] and exponential-weighting variants; and actively adaptive methods that detect distribution changes, of which AdSwitch [7] is a representative with regret guarantees under an unknown number of changes. Under milder, more informative statistical assumptions about the regimes, Bayesian approaches such as Thompson sampling [8, 9], possibly in a contextual form, may exploit structure that the worst-case methods cannot. The appropriate choice depends on the regime-switching statistics and the separation of the per-arm reward distributions; quantifying the latter is one purpose of Section 5.

3 Simulation framework

3.1 Time-domain plant model

We use a time-domain simulator ("Lightsaber") of the pitch degree of freedom of a single arm cavity, run as a per-sample closed loop at a sampling frequency $f_s = 256\,\text{Hz}$. The principal parameters are an arm length $L = 3994.5\,\text{m}$, radii of curvature $R_1 = 1934\,\text{m}$ and $R_2 = 2245\,\text{m}$, ITM transmissivity $T_1 = 0.014$, laser wavelength $\lambda = 1064\,\text{nm}$, and a nominal arm-cavity input power $P_{\text{in}} = 705\,\text{W}$ (corresponding to a nominal circulating power of order 200 kW).

At each time step the plant (i) injects test-mass pitch noise obtained by filtering colored seismic and damping-loop (OSEM) noise through measured suspension transfer functions; (ii)

Table 1: Regime definitions: multiplicative scale factors applied to the nominal configuration.

	R_0 (nominal)	R_1 (high-micro)	R_2 (extreme)
Optical-power scale	1.00	0.95	0.90
Sensing-noise scale	120	12	0.25
Seismic-input scale	1.0	6.0	20.0
OSEM (damping) scale	1.0	1.0	1.0

computes the cavity power from Eq. (3); (iii) forms the radiation-pressure torques on the two masses,

$$\tau = \frac{2}{c}\Big(P_{\mathrm{cav}}\,\mathbf{s} + \mathbf{s}_0\,(P_{\mathrm{cav}} - P_{\mathrm{dc}})\Big), \tag{4}$$

with $\mathbf{s}$ the beam-spot vector, $\mathbf{s}_0$ a static beam-spot offset defining the operating point, and P_{dc} a reference power; (iv) propagates torque and actuation to mirror angle through the radiation-pressure (TM→P) and actuator (PUM→TM) state-space models; (v) reads out the two eigenmodes via Eq. (1) with additive sensing noise; (vi) applies a power-dependent Sidles–Sigg compensation path; and (vii) computes the controller actuation. The controllers act on the eigenmode error and their output is mapped back to the local actuation basis. All recursive elements (plant, compensator, controller) are second-order-section (SOS) digital filters with persistent internal state, so that the simulation is a faithful continuous closed loop rather than a sequence of independent batches.

3.2 Validation against the reference simulator

The plant used here is a refactored version of an earlier, component-graph reference implementation. We verified that the refactoring preserves the physics. The two core plant transfer functions—the actuator response (PUM→TM, gain 93.53) and the radiation-pressure response (TM→P, gain 2.5677)—are identical zero–pole–gain models; the three measured suspension transfer functions are bit-identical between the two codebases (maximum amplitude difference 0); and the high-pass, Fabry–Perot, eigenmode and beam-spot expressions coincide. The differences between the two implementations are confined to the deliberately modified experimental layer—the static operating point $\mathbf{s}_0$, the form of the Sidles–Sigg compensator, the controller family (Section 4), and the regime-dependent noise inputs—and none constitutes a change to the underlying plant. We therefore treat the present plant as physically equivalent to the reference. **[TODO: Optionally add a quantitative dynamic cross-check (matched-configuration output PSD overlay).]**

4 Problem formulation

4.1 Operating regimes

We define three operating regimes that differ along physically motivated axes: the optical power (and hence the Sidles–Sigg coupling strength), the seismic input level, and the sensing (readout) noise. The scale factors, applied to the nominal configuration, are listed in Table 1; the underlying amplitude spectral densities are derived from O4-era data. Regime R_0 is the nominal, low-ground-motion state; R_1 a high-microseism state; and R_2 an extreme state with strongly elevated seismic input but a comparatively clean sensor.

4.2 Controller family

We consider three fixed hard-mode controllers, C_0, C_1, C_2, which share an identical filter shape and differ only by an overall DC gain (30, 44.2, 50 respectively). They are therefore a oneparameter gain family: driven by the same error signal, their outputs are proportional, $u^{(j)} = (g_j/g_i)\,u^{(i)}$. The goal is to *select* among them online, not to retune them. The soft-mode controller is common to all three and is not varied.

4.3 Observables and scalar reward

Performance is assessed on the hard eigenmode only, through two observables: the hard-mode error signal e_h (the controller input/readout) and the hard-mode control effort u_h (the controller output, transformed to the eigenbasis). A linear scalarisation of error and control, $J = \alpha\,\mathrm{RMS}(e_\mathrm{h}) + \beta\,\mathrm{RMS}(u_\mathrm{h})$, cannot yield a consistent regime–controller preference, because different regimes call for opposite error/control trade-offs; a fixed trade-off slope therefore fails. We instead use a multiplicative (logarithmic) cost. In its minimal form,

$$J = \rho\,\log\frac{\mathrm{RMS}(e_\mathrm{h})}{\min_C \mathrm{RMS}(e_\mathrm{h})} + \log\frac{\mathrm{RMS}_{10\text{–}30\,\mathrm{Hz}}(u_\mathrm{h})}{\min_C \mathrm{RMS}_{10\text{–}30\,\mathrm{Hz}}(u_\mathrm{h})}, \tag{5}$$

with a single global coefficient $\rho \approx 1.095$, which already yields the intended one-to-one mapping. In the implementation used for the bandit reward we adopt a generalised, multi-band form: the error and control signals are decomposed into a set of Butterworth frequency bands, and the reward is a weighted sum of the log band-powers evaluated on 2 s blocks,

$$Z = \sum_i w_i \log b_i, \qquad r_\mathrm{raw} = -Z, \tag{6}$$

with band features b_i (RMS per band of e_h and u_h) and fixed weights w_i calibrated so as to reproduce the designed mapping. Because the downstream policy expects a bounded reward, the per-segment mean raw score is mapped to $[0, 1]$ by a logistic,

$$r = \left[1 + \exp\bigl(-(\bar{r}_\mathrm{raw} - r_0)/\sigma_0\bigr)\right]^{-1}, \qquad r_0 = 167.5,\ \sigma_0 = 2.5. \tag{7}$$

The implications of this normalisation are examined in Section 5.

4.4 Designed regime–controller mapping

The regimes and controllers are constructed so that, under the cost of Eq. (5)–(6), each regime has a distinct optimal controller: $R_0 \to C_0$, $R_1 \to C_1$, $R_2 \to C_2$. Physically, the noisy sensor of R_0 favours the low-gain controller (which injects less control noise), while the clean sensor and elevated seismic input of R_2 favour the high-gain controller (which rejects the larger disturbance). The accuracy of this mapping is the subject of Section 5.

5 Characterisation of the reward landscape

We characterise the reward of every controller in every regime by simulating all nine (regime, controller) combinations. The band features are obtained by filtering each error/control signal *continuously* (the band-pass filter state is carried across the run; the lowest band, 0.05–0.3 Hz, carries the largest weight and requires this), and the reward of Eq. (6) is the mean over 2 s blocks. Statistics are accumulated over five independent noise realisations of 360 s each (a 60 s filter warm-up discarded; ∼750 settled blocks per cell). Table 2 reports the mean raw and logistic-mapped reward; Fig. 1 shows the per-controller distributions and means.

Three features stand out. First, the designed one-to-one mapping holds in *all three* regimes: $R_0 \to C_0$, $R_1 \to C_1$ and $R_2 \to C_2$ are each the per-regime optimum (Table 2). Second, the separation is regime-dependent. It is large and highly robust in the nominal regime R_0 (raw gap 0.84, $\sim 35\sigma$ over noise realisations) and robust in the intermediate regime R_1 (gap 0.16, $\sim 4.5\sigma$), but marginal in the extreme regime R_2, where C_1 and C_2 are nearly equivalent (gap

Table 2: Mean reward per (regime, controller). Raw score r_{raw} [Eq. (6)] and logistic (bandit-space) reward [Eq. (7)]. Bold marks the per-regime optimum.

	raw r_{raw}			logistic r		
	C_0	C_1	C_2	C_0	C_1	C_2
R_0	**168.59**	167.75	167.38	**0.597**	0.521	0.487
R_1	165.52	**165.68**	165.39	0.325	**0.338**	0.314
R_2	167.43	168.23	**168.28**	0.492	0.565	**0.569**

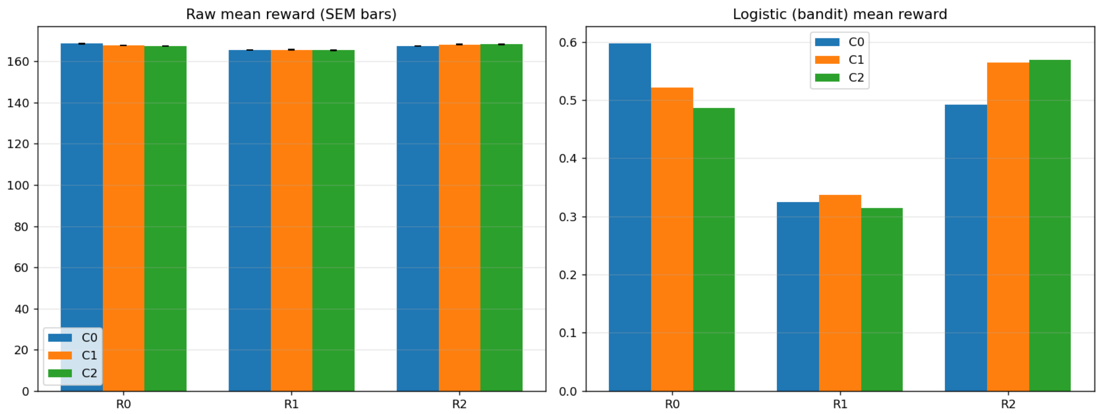

Figure 1: Mean reward per regime and controller, in the raw score (left, with standard-error bars) and in the logistic bandit-reward space (right). The diagonal (regime-optimal) controller is the largest in each regime, and the logistic reward is non-saturating (all values 0.31–0.60).

0.05, $\sim 0.8\sigma$). Because C_1 and C_2 are close in R_2, the per-step regret of confusing them is small and contributes little to the integrated cost. Third, with the corrected continuous-filter reward the logistic normalisation of Eq. (7), centred at $r_0 = 167.5$, is *non-saturating* over the reward range ($\approx$ 165–169): the bandit-space rewards span 0.31–0.60 across all regimes (Fig. 1, right), so the per-regime separation is preserved rather than compressed. A learnable signal is therefore present in every regime—strongest in the nominal regime, weakest in the extreme regime—which is what makes the online-selection problem well posed.

6 Dynamics of controller switching

A controller switch perturbs the loop transiently. To quantify this cost in isolation from the (separate) steady-state reward difference between controllers, we use a common-random-numbers design. For a transition $C_i \to C_j$ we run a *switch* simulation (C_i for a warm-up T_w, then C_j) and an *always-C_j* reference on the *identical* noise realisation, and form the post-switch reward difference

$$\Delta r(t) = r_{\text{switch}}(t) - r_{\text{always-}C_j}(t). \tag{8}$$

After the switch both runs apply C_j on the same noise, so the steady-state reward of C_j cancels and Δr isolates the transient; that $\Delta r(t) \to 0$ at long times (Fig. 2) confirms the cancellation. We report the dip depth $\min_t \Delta r$ and the integrated loss $\int \Delta r \, \mathrm{d}t$, averaged over six seeds, with $T_\text{w} = 64\,\text{s}$ (long enough for the continuous reward filters to settle) and a 40 s test window. All runs use the validated fast kernel and the corrected continuous-filter reward.

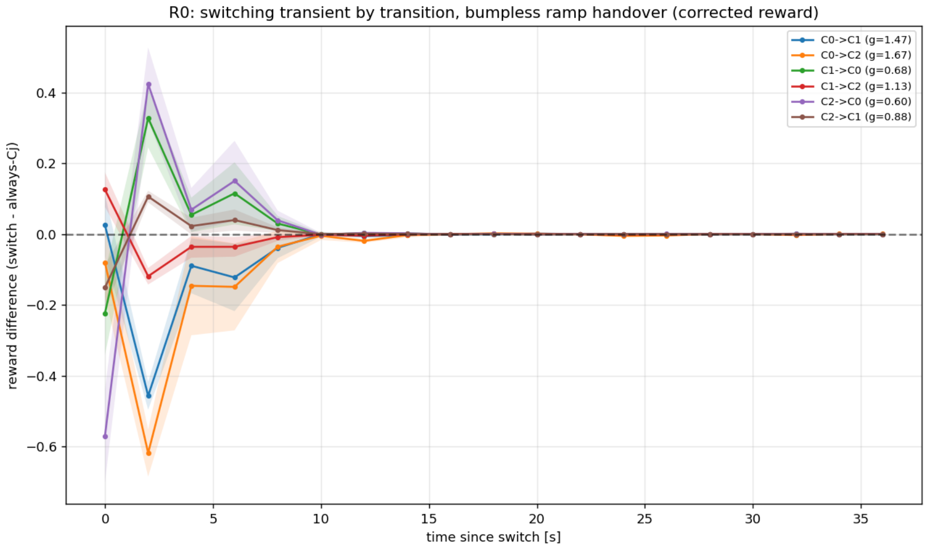

Figure 2: Switching transient $\Delta r(t)$ in R_0 for all six transitions, with the realistic bumpless-ramp handover. The dip is shallow ($\lesssim 0.6$ in reward) and recovers within a few seconds for every transition, so the switch cost is small under a proper handover.

6.1 Transient cost and its asymmetry

A switch produces a brief reward dip; the loop remains stable throughout (no divergence). With the *naive* handovers the cost is strongly asymmetric in the direction of the gain change: gain-*decreasing* switches (from a higher- to a lower-gain controller) are far costlier than gain-increasing ones, the extreme case in R_0 being $C_2 \to C_0$ (the largest gain drop), which reflects the abrupt re-equilibration of the loop when the effective gain changes discontinuously. With the realistic handovers this asymmetry is largely removed (Fig. 2): every transition incurs only a shallow, few-second dip.

6.2 Dependence on the handover implementation

The magnitude of the transient depends far more on *how* the controller is handed over than on the transition itself. We compare four implementations of the switch:

- **naive carry-over ("hot")**: the outgoing controller's internal filter state is transferred verbatim into the incoming controller. Because the gain difference resides in an internal filter section, the carried state is mis-scaled and the transient is *worsened*.
- **state reset ("cold")**: the incoming controller's filter state is zeroed. In a digital control system this produces an output discontinuity and an actuator impulse, and is not representative of switching a loop that must remain locked.
- **parallel banks**: all candidate controllers run continuously on the same error, each maintaining its own state; only the active output drives the suspensions. A switch changes which output is applied, producing a clean (g_j/g_i) step in actuation.
- **bumpless ramp**: as parallel banks, but the applied command is cross-faded between the outgoing and incoming controller outputs over a short window, $u(t) = (1-\alpha)\, u^{(i)}(t) + \alpha\, u^{(j)}(t)$ with $\alpha : 0 \to 1$. For this gain family the blend is exactly a continuous slew of the effective loop gain from g_i to g_j, and is stable provided the loop has gain margin across the range. This models the gain-ramped, parallel-filter-bank handover used in detector digital-control systems.

Table 3: Switching cost for $C_2 \to C_0$ in R_0 (gain ratio 0.60) by handover implementation. Integrated loss in (raw score)·s; dip in raw score.

Handover	integrated loss	dip depth	realism
naive carry-over (hot)	-140	-20.3	implementation artifact
state reset (cold)	-8.4	-3.3	actuator impulse; unrealistic in lock
parallel banks	-5.1	-2.2	realistic; gain-step
bumpless ramp, 8 s	-2.8	-1.1	realistic; smooth
bumpless ramp, 2 s	$\mathbf{+0.2}$	$\mathbf{-0.6}$	realistic; near-eliminated

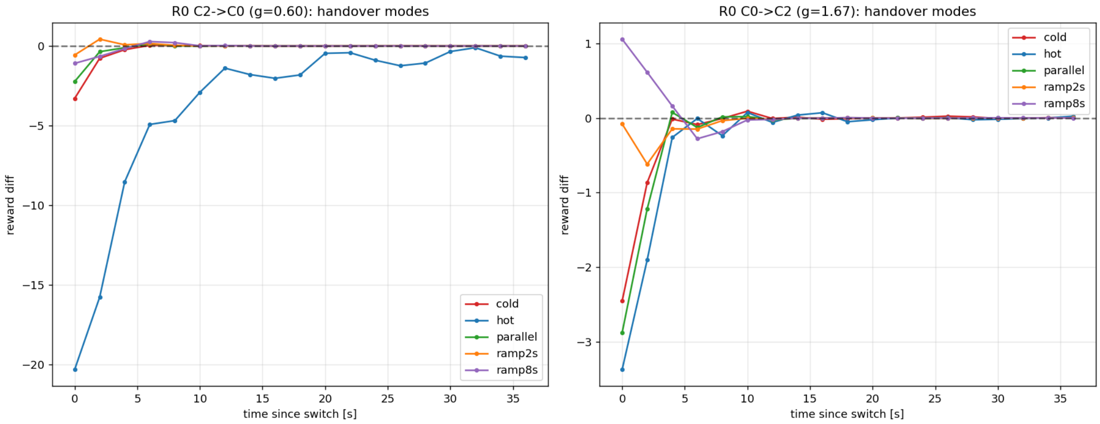

Figure 3: Switching transient by handover implementation, for $C_2 \to C_0$ (left) and $C_0 \to C_2$ (right) in R_0. A bumpless ramp (orange/purple) suppresses the transient that the naive carry-over (blue) and, to a lesser extent, the state reset (red) and parallel-bank (green) handovers exhibit.

Table 3 and Fig. 3 compare these on the worst relevant case, $C_2 \to C_0$ (a switch into the R_0 optimum). The integrated loss spans more than two orders of magnitude: the naive carry-over is by far the worst, state reset and parallel banks are comparable and moderate, and a short (~ 2 s) bumpless ramp very nearly eliminates the transient.

The central conclusion is that the apparent switching penalty is, for this controller family, largely an artifact of the handover *implementation* rather than an intrinsic physical cost. With a bumpless, parallel-bank handover the per-switch transient (integrated loss $\lesssim 1$ in reward·s; a few-second dip of $\lesssim 0.6$) is small: averaged over a hold window of hundreds of seconds it shifts the window-mean reward by far less than the per-regime separation between controllers (Section 5), so an online policy can switch freely. Two qualifications apply. First, this relies on the controllers forming a gain family, for which the cross-fade is provably a smooth gain slew; for controllers of different shape—e.g. future learned controllers—a convex blend of two stabilising controllers is not guaranteed to be stabilising mid-ramp, and the switching cost must be re-measured. Second, even an ideal handover cannot remove the physical re-equilibration of the loop to a different closed-loop bandwidth; the recovery times of a few seconds reflect this irreducible component.

7 Online controller selection

[TODO: This section reports the bandit experiments and is not yet complete.]

7.1 Algorithms

We compare an actively adaptive, distribution-change-detecting policy (AdSwitch [7]), which makes no assumptions about the regime-switching statistics, against (i) a Bayesian policy exploiting milder statistical structure (Thompson sampling, possibly contextual) [8, 9]; (ii) passively forgetting baselines (discounted and sliding-window UCB) [6]; and the reference baselines (iii) each fixed controller, (iv) an oracle that always plays the per-regime optimum, and (v) a simple forecast-driven rule mimicking current operational practice. **[TODO: finalise the algorithm set and the choice of the main method.]**

7.2 Experimental protocol

The environment evolves *adiabatically* between regimes (slow ramps and holds rather than abrupt steps); the controller is held fixed within each decision interval of duration T_{hold} and the bandit is updated with the segment-mean reward; switching uses the bumpless handover of Section 6.2. The figure of merit is the cumulative (time-integrated) reward over long horizons, and the per-step regret against the time-varying optimum, decomposed into wrong-arm and switching-cost contributions. **[TODO: specify T_{hold}, horizon, the non-stationarity model and its parameters, and the number of repetitions.]**

7.3 Results

[TODO: Cumulative-reward curves and regret table for all policies; sensitivity to T_{hold} and to the regime-switching rate.]

8 Discussion

The infrastructure and measurements above delimit what an online controller- selection policy can and cannot achieve in this setting. The regime–controller problem is well posed, with the one-to-one mapping holding in all three regimes and a non-saturating reward; the per-regime separation is largest in the nominal regime and smallest (though still favouring the designed controller) in the extreme regime, so the achievable advantage of a selection policy is modest and accrues across all regimes. At the same time, with a realistic bumpless handover the cost of switching is small, so regret is governed primarily by *which* controller is chosen rather than by the act of switching. These two observations frame the bandit comparison of Section 7: the relevant question is how quickly and reliably a policy identifies the regime-optimal controller as the environment drifts, with switching essentially free. **[TODO: expand once bandit results are in; comment on algorithm choice and on the path to learned (NN) controllers, for which the reward separation is expected to be larger and the switching cost must be re-assessed.]**

9 Conclusion

[TODO: Summarise contributions and the bandit outcome; restate the headline that switching is nearly free under a proper handover, and the implications for autonomous alignment control.]

Acknowledgements

[TODO: funding and computing acknowledgements.]

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[10] **[TODO: replace with a citable reference (or remove) for the adiabatic-non-stationarity argument.]**